Lecture 2 2025-09-02

Derivative: $f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$

$f^{(n)}(x_0)$ = n'th derivative at x_0

Thm 1.6 If f is differentiable at x_0, then f is continuous at x_0.

Notation: $C^r([a,b]) = \{$ r times continuously differentiable functions defined on interval $[a,b]$ $\}$

Also works for $r = \infty$: infinitely often differentiable/smooth functions

Examples: $e^x \in C^\infty(\mathbb{R})$

$\sin x \in C^\infty(\mathbb{R})$

$\ln x, \log x \in C^\infty((0,\infty))$

$|x| \in C(\mathbb{R})$ & differentiable (but not continuously)

But $|x| \in C^\infty([a,b])$ if $0 \notin [a,b]$

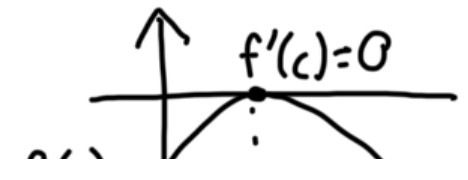
was a typo. Should be if $0 \notin [a,b]$!

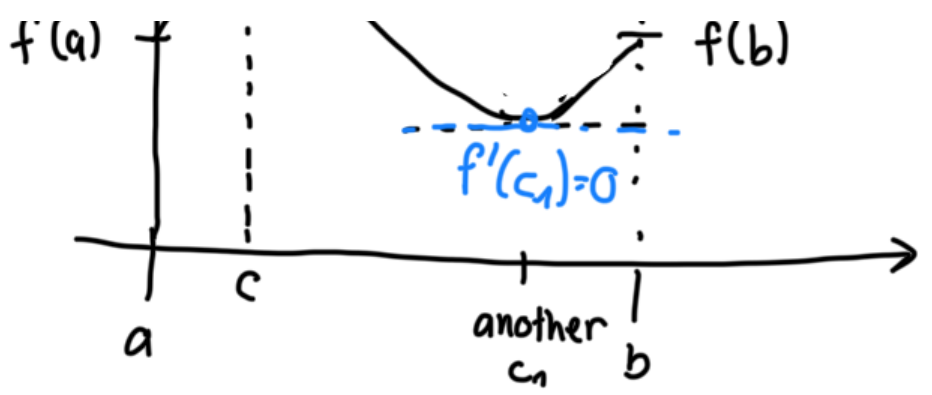
Some derivative theorems (useful for deriving error estimates)

Thm 1.7 (Rolle's theorem)

Suppose $f \in C([a,b])$ and f is differentiable on (a,b).

If $f(a) = f(b)$, there exists $c \in (a,b)$ such that $f'(c) = 0$.

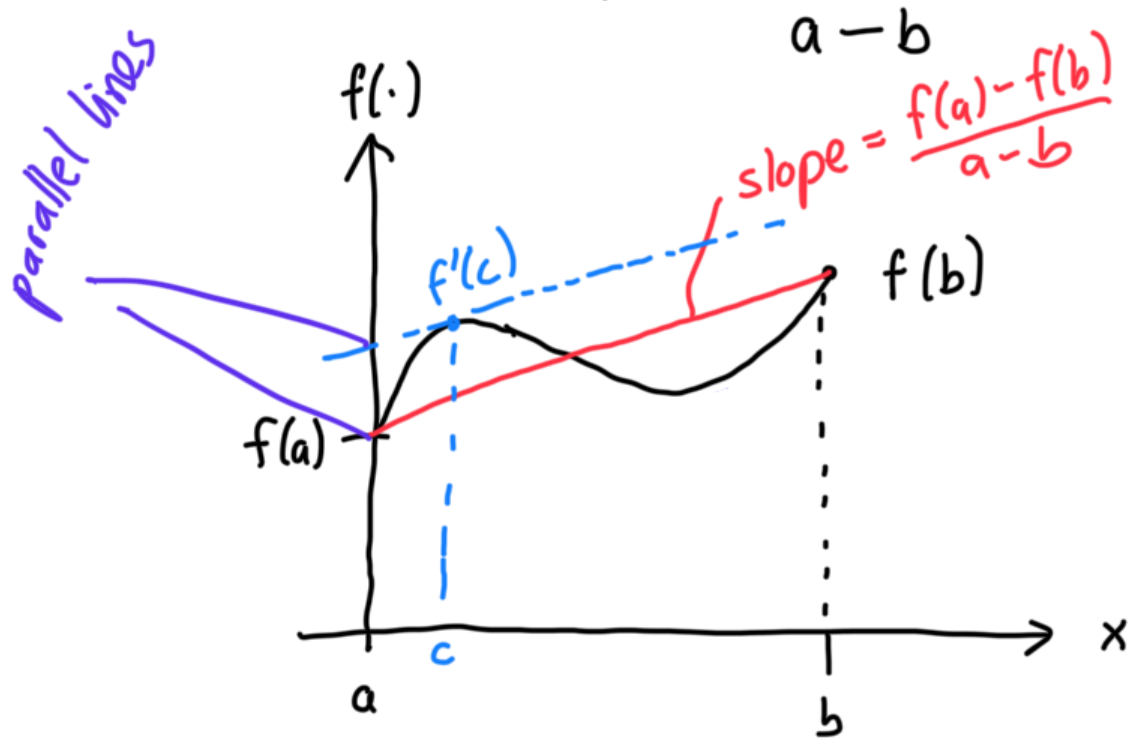




Thm 1.8. (Mean Value Theorem MVT)

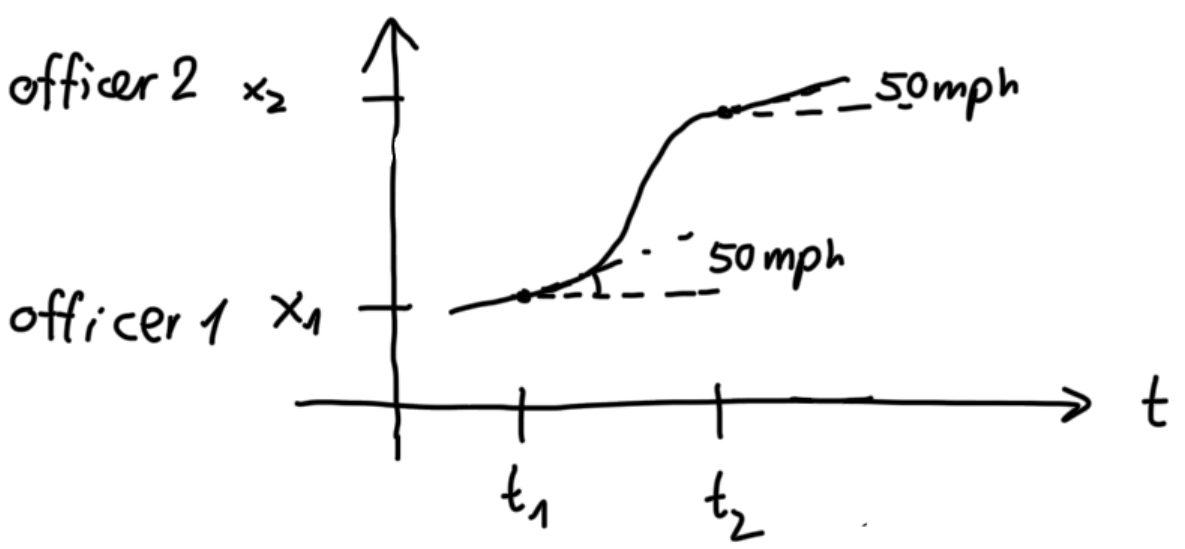
If $f \in C([a,b])$ and f is differentiable on (a,b), then there is a number $c \in (a,b)$ such that

$$f'(c) = \frac{f(a)-f(b)}{a-b}$$



Note: Rolle's theorem is a special case of the MVT

Application: Police can prove that you were speeding without a radar gun



If you show up at x_2 at time

$t_2 < t_1 + \frac{x_2 - x_1}{\text{speed limit}}$ then there was at least one point in (x_1, x_2) where you were speeding

$$t_2 - t_1 < \frac{x_2 - x_1}{\text{speed limit}}$$

$$\iff \text{speed limit} < \frac{x_2 - x_1}{t_2 - b_1} = \text{average speed} = \text{instantenuous speed at some time } t^* \in (t_1, t_2)$$

↑ by MVT

Thm 1.9 Extreme Value Theorem (EVT)

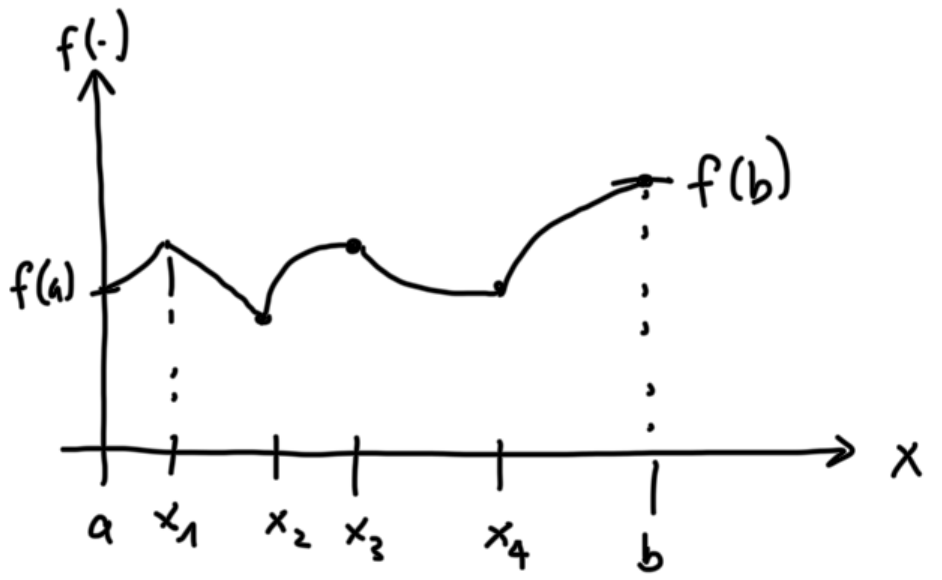
If $f \in C([a,b])$ then $\exists\ c_1, c_2 \in [a,b]$ such that

$$\underbrace{f(c_1)}_{\min} \leq f(x) \leq \underbrace{f(c_2)}_{\max} \quad \text{for all } x \in [a,b]$$

So f attains its minimum and maximum if it is continuous.

In addition, if f is differentiable on (a,b) then the c_1 and c_2 are either at the endpoints of $[a,b]$ or where $\underbrace{f'(x) = 0}_{\text{critical point}}$.

Corollary: If $f \in C([a,b])$ and f is piecewise differentiable (PWD), i.e. $f'(x)$ exists for (x_{i-1}, x_i), $1 \leq i \leq n$ ($n < \infty$), where $a = x_0 < x_1 < x_2 < \ldots < x_n = b$ then c_1 and c_2 occur either at a, b, one of the x_i $(i = 1, 2, \ldots, n-1)$ or where $f'(x) = 0$



"Proof": Apply EVT to each subinterval $[x_{i-1}, x_i]$ and then take min/max over set of extrema from each subinterval.

↑ piecewise differentiable fcn as defined above

"Algorithm": To find the extrema of $f \in C([a,b]) \cap PWD([a,b])$

compare the values of $f(x)$ at

- the endpoints $x=a$, $x=b$
- critical points $x=c$ where $f'(c)=0$ or
- where $f'(x)$ does not exist

Example: Find the extrema of $f(x) = xe^x$ on $[0,2]$

Solution: Use "algorithm"

- Evaluate f at endpoints:

$f(a) = f(0) = 0$

$f(b) = f(2) = 2e^2$

- Evaluate f at critical points:

$f'(x) = e^x + xe^x = (1+x)e^x \overset{!}{=} 0$

does not happen in $[0,2]$
since $e^x > 0$ and $1+x > 0$

$\Rightarrow$ no critical points in $[0,2]$

$\Rightarrow$ min is at $a=0$ & $=0$

max is at $b=2$ & $=2e^2$

Example 2: $f(x) = xe^{-x}$ on $[0,2]$

- Evaluate f at endpoints:

$f(0) = 0$

$f(2) = 2e^{-2}$

- Evaluate f at critical points:

$f'(x) = e^{-x} - xe^{-x} = (1-x)\underbrace{e^{-x}}_{>0} \overset{!}{=} 0$ on $[0,2]$

$\Rightarrow$ $x=1$ is the only critical point ($f'(1)=0$)

$f(1) = e^{-1}$

$$f(2) = \frac{2}{e}e^{-1} < e^{-1} = f(1)$$

($\frac{2}{e}<1$)

$\Rightarrow$ min at $x=0$ and $=0$

max at $x=1$ and $=e^{-1}$

Thm 1.11 *Intermediate Value Theorem (IVT)*

If $f \in C([a,b])$ and K is between $f(a)$ and $f(b)$ then $\exists\, c \in (a,b)$ such that $f(c)=K$

Example: Show that $f(x)=x^3+2x+k$ crosses the x-axis exactly once, regardless of the value of k.

Solution: If $k=0$ then $c=0$ works: $f(0)=0$

Otherwise consider

$f(k) = k^3+3k$

$f(-k) = -k^3-k$

} have opposite signs

$\Rightarrow$ by IVT $\exists\, c \in (-|k|, |k|)$ s.t. $f(c)=0$

Uniqueness: $f'(x) = 3x^2+2 > 0$ for all x.

$\Rightarrow$ f is strictly increasing.

If f crossed the x-axis at two points c_1 and c_2 so $f(c_1)=f(c_2)=0$ then by Rolle's theorem there exists $c \in (c_1,c_2)$ such that $f'(c)=0$ which contradicts $f'(x)>0$. $\Rightarrow$ f crosses the

x-axis at most once.

Proposition: Suppose $f \in C([a,b])$ and $g(x) = |f(x)|$.
Then $g \in C([a,b])$ and the extrema of g occur at extrema or zeros of f.

Proof of continuity:

$$|g(x) - g(x_0)| = ||f(x)| - |f(x_0)||$$

$$\leq |f(x) - f(x_0)| \longrightarrow 0 \quad \text{as } x \to x_0$$

$\uparrow$ triangle inequality